

Suresh Kumar Amalapuram, Ph.D

CONTACT INFORMATION

Assistant Professor (Grade-II),
School of Artificial Intelligence & Data Engineering,
Indian Institute of Technology Ropar,
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RESEARCH INTERESTS

Machine learning and its application to security

- Continual learning, Intrusion detection
- Tailored ML techniques for security and system

EDUCATION

Ph.D. in Computer Science & Engineering [July 2019 to Sept 2024]
Dept. of Computer Science & Engineering
Indian Institute of Technology Hyderabad (CGPA: 8/10)
Advisors: Dr. Bheemarjuna Reddy Tamma, Dr. Sumohana S Channappayya
Thesis: Continual Learning for Intrusion Detection Systems: Addressing Concept Drift in Network Traffic

M.Tech in Information Technology [July 2014 to May 2016]
Dept. of Computer Science & Engineering
Indian Institute of Technology Kharagpur (CGPA: 8.2/10)

B.Tech in Information Science & Technology [July 2007 to May 2011]
Acharya Nagarjuna University, Guntur (Percentage: 77/100)

EMPLOYMENT

Assistant Professor (Grade-II) [March 2026 to Current]
School of Artificial Intelligence & Data Engineering
Indian Institute of Technology Ropar

Postdoctoral Researcher [Sept 2025 to March 2026]
Advisor: Prof. Bheemarjuna Reddy Tamma and Prof. Mahesh K. Marina
Indian Institute of Technology Hyderabad

Postdoctoral Researcher [Jan 2025 to July 2025]
Advisor: Prof. Mahesh K. Marina
University of Edinburgh

Postdoctoral Researcher [Aug 2024 to Jan 2025]
Advisor: Prof. Bheemarjuna Reddy Tamma
Indian Institute of Technology Hyderabad

Software Development Engineer - 2 [July 2016 to April 2019]
Oracle India Private Limited, Bengaluru

Assistant Manager [June 2011 to May 2014]
Family owned business

PUBLICATIONS

- **S. K. Amalapuram**, B. Shrestha, C.S.R. Murthy, B. R. Tamma, and S. S. Channappayya, "Semi-supervised continual malware detection: Tackling concept drift on a budget" in *ACM Transactions on AI Security and Privacy* ((**ACM TAISP**), [\[link to paper\]](#) Accepted, June 2026
- M.A. Haque, I. Hossain, M. Kamol, J. Alam, **S. K. Amalapuram**, S. Talukder, M. S. Rahman, "LAMDA: A Longitudinal Android Malware Benchmark for Concept Drift Analysis" in *Proceedings of the 14th International Conference on Learning Representations (ICLR)*, Rio de Janeiro (Brazil), 2026. [\[link to paper\]](#) (Core A*) [citations: 11]
- H. Patel, **S. K. Amalapuram**, B.R. Tamma, "IoT - FedMalDetect: Federated Learning based Malware Detection for IoT Edge Devices" in *Proceedings of the 15th IEEE Conference on Communications and Network Security (IEEE CNS)*, pp. 1–9, Avignon (France) 2025. [\[link to paper\]](#)
- **S. K. Amalapuram**, B. R. Tamma, S. S. Channappayya "Experimental Insights on continual learning for network intrusion detection" in *Proceedings of the IEEE International Conference on Data Mining Workshops (ICDMW)*, pp. 1551-1557, Washington (USA), 2025. [\[link to paper\]](#)
- **S. K. Amalapuram**, B. R. Tamma, and S. S. Channappayya, "Spider: A semi-supervised continual learning-based network intrusion detection system," in *Proceedings of the 43rd IEEE Conference on Computer Communications (IEEE INFOCOM)*, pp. 571–580, Vancouver (Canada), 2024. [\[link to paper\]](#) (Core A*) [citations: 33]
- **S. K. Amalapuram**, S. S. Channappayya, and B. R. Tamma, "Augmented memory replay-based continual learning approaches for network intrusion detection," in *Proceedings of the 37th Conference Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, pp. 17156–17169, New Orleans (USA), 2023. [\[link to paper\]](#) (Core A*) [citations: 53]
- **S. K. Amalapuram**, A. Tadwai, R. Vinta, S. S. Channappayya, and B. R. Tamma, "Continual learning for anomaly based network intrusion detection," in *Proceedings of the 14th International Conference on COMMunication Systems & NETWORKS (COMSNETS)*, pp. 497–505, Bangalore (India), 2022. [\[link to paper\]](#) [citations: 64]
- **S. K. Amalapuram**, T. T. Reddy, S. S. Channappayya, and B. R. Tamma, "On handling class imbalance in continual learning based network intrusion detection systems," in *Proceedings of the 1st ACM International Conference on AI-ML-Systems*, pp. 70–77, Bangalore (India), 2021. [\[link to paper\]](#) [citations: 19]

UNDER REVIEW

- **S. K. Amalapuram**, S. Kumar, B. R. Tamma, and S. S. Channappayya, "SOUL: A Semi-supervised Open-world continUal Learning method for network intrusion detection" (*ACM Transactions on AI Security and Privacy*) [citations: 6]
- M. Khoja, **S. K. Amalapuram**, M. Marina, "RKS-Net: RAN KPIs Summarization Neural Network" (at ACM CONEXT 2026).
- M.A. Haque, I. Hossain, **S. K. Amalapuram**, M. S. Rahman, "CITADEL: A Semi-Supervised Active Learning Framework for Malware Detection Under Continuous Distribution Drift" (ICSE 2027)

AWARDS & GRANTS

- Lead a team (of three), that won 1st prize (INR 100000) in malware analysis hackathon 2024 at IIT Madras
- Google travel grant to attend NeurIPS 2023
- NeurIPS 2023 scholar award
- COMSNETS 2024 international travel award by LRN foundation
- ACM travel grant for IEEE INFOCOM 2024
- PhD fellowship grant from MoE, Govt. of India for the period (2019-2024).

SERVICE

- **Reviewing** - ICDCS 2025, NeurIPS 2025, ICLR 2026, ACM Mobicom 2026, IEEE Communication Letters, Computers & Security, ICVGIP, Complex & Intelligent Systems
- **Mentoring** - Supervised UG and PG students at IIT Hyderabad
- **Teaching Assistant** - Introduction to Programming, Network Security, Deep Learning

PROGRAMMING

Pytorch, Python, Java, .NET, C/C++, L^AT_EX, etc.

LANGUAGE SKILLS

Telugu (Native), **English** (Read, Write, Speak), **Hindi** (Read, Write, Speak)

PERSONAL INFORMATION

Permanent Address: 27-122, Ramalayam Street, Bapatla district
Karamchedu - 523168, Andhra Pradesh, India
Nationality: Indian

REFERENCES

Dr. Bheemarjuna Reddy Tamma
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CS-308
Dept. of Computer Science & Engineering
Indian Institute of Technology Hyderabad
Kandi, Sangareddy, Telangana,
India, 502285
E-mail: tbr@cse.iith.ac.in

Dr. Sumohana S. Channappayya
Professor
EE-605,
Dept. of Electrical Engineering
Indian Institute of Technology Hyderabad
Kandi, Sangareddy, Telangana,
India, 502285
E-mail: sumohana@ee.iith.ac.in